

Intelligent System for Plant Disease Detection and Remedy Recommendation

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INTRODUCTION

Agriculture is at risk of huge losses if plant disease detection is delayed or not done properly. Manual methods are laborious, unreliable, and require experts, who are most of the time not present in the rural areas. Leaf image-based deep learning models including U-Net, ResNet50, ViT, VGG16, VGG19, AlexNet, DenseNet, and EfficientNetB4 have made possible the detection that is both accurate and automated. The project offers a web-based platform that gives on the spot disease diagnosis and treatment guidelines.



PROBLEM STATEMENT

Plant disease detection currently relies on manual inspection, which is slow, labor-intensive, and dependent on expert knowledge that is not always available. Farmers often misidentify symptoms or detect diseases at a late stage, leading to significant crop losses. Additionally, existing computer-based solutions are either too basic or lack sufficient accuracy, making them ineffective for handling multiple crops and diverse disease types.

OBJECTIVES

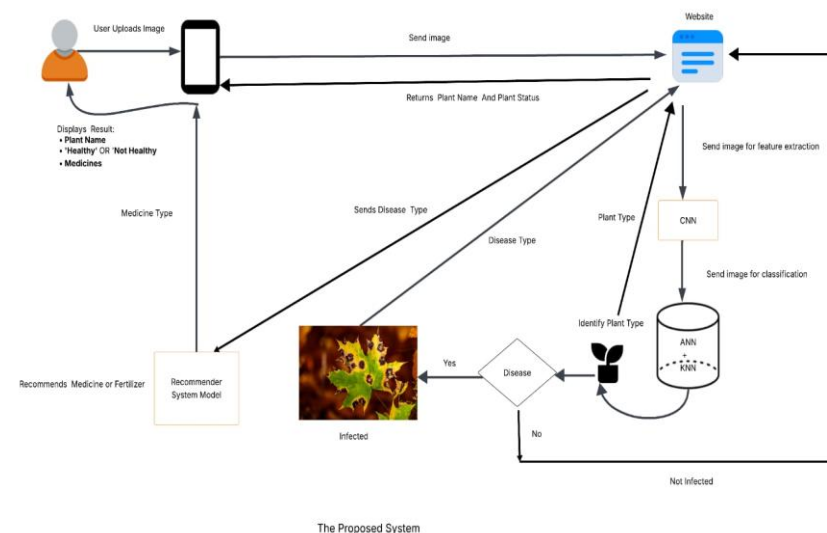
The objective of this project is to develop a system for **automatic plant disease detection** using leaf images. It uses deep learning models such as **U-Net, ResNet50, ViT, VGG16, VGG19, AlexNet, DenseNet, and EfficientNetB4** for accurate classification of healthy and diseased plants. The system aims to support **early detection and reduce crop losses**.

METHODOLOGY

The system begins with **leaf image input**, either from a dataset or uploaded by the user. The image undergoes **preprocessing** steps such as resizing (224×224), normalization, and data augmentation to improve model performance.

The processed image is then passed through multiple **deep learning models** including **U-Net, ResNet50, Vision Transformer (ViT), VGG16, VGG19, AlexNet, DenseNet, and EfficientNetB4** for accurate classification. The system predicts whether the leaf is healthy or identifies the disease type.

Finally, the result is displayed on a **web-based interface**, along with **recommended remedies**, enabling real-time and user-friendly access.

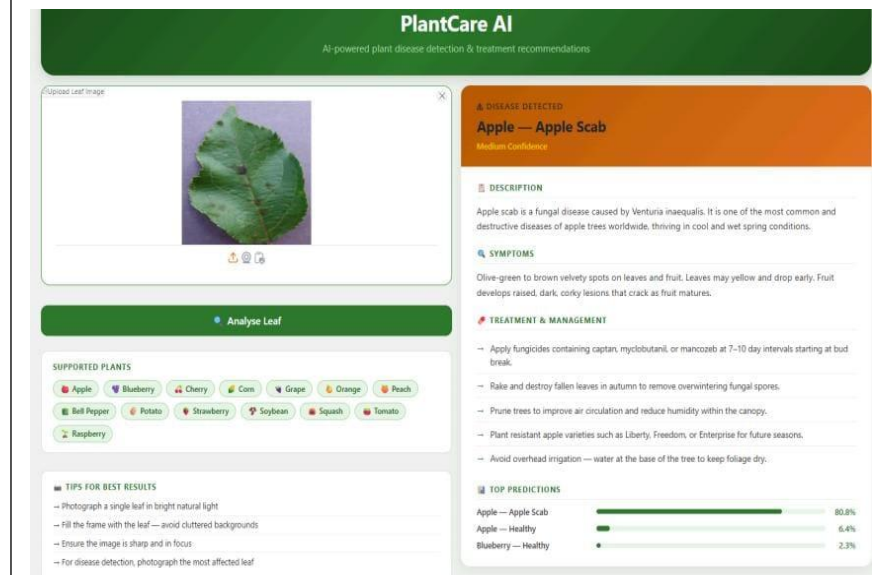


IMPLEMENTATION

The system is developed using **Python** with deep learning libraries like **TensorFlow** and **Keras** for training and testing multiple models for plant disease detection. Image preprocessing and data handling are performed using **NumPy** and **Pandas**, while model evaluation is done using standard metrics. A **Flask-based backend** handles user requests and processes uploaded leaf images, while the frontend built with **HTML, CSS, and JavaScript** provides an interactive interface. The system predicts the disease and displays results along with **remedy recommendations in real time**, ensuring a fast and reliable solution.

RESULTS

The system successfully provides real-time predictions through a web-based interface. Performance evaluation was carried out using accuracy, loss graphs, and confusion matrix.



CONCLUSION AND FUTURE SCOPE

Deep learning enables **accurate and automated plant disease detection**, providing real-time predictions and helping reduce crop losses. Future Work:

- Develop a **robust backend system** to handle user requests
- Create a **database for storing and managing remedy information**.
- Develop a **multilingual chatbot interface** to provide remedy guidance in regional languages.
- Deploy the system on a **cloud platform for online access**.

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